

23-4a A Numerical Example

Table 2 shows some data for an economy that produces only two goods: hot dogs and hamburgers. The table shows the prices and quantities produced of the two goods in the years 2019, 2020, and 2021.

Table 2

Real and Nominal GDP

This table shows how to calculate real GDP, nominal GDP, and the GDP deflator for a hypothetical economy that produces only hot dogs and hamburgers.

Prices and Quantities

Year	Price of Hot Dogs	Quantity of Hot Dogs	Price of Hamburgers	Quantity of Hamburgers
2019	\$1	100	\$2	50
2020	2	150	3	100
2021	3	200	4	150

Calculating Nominal GDP

2019	$(\$1 \text{ per hot dog} \times 100 \text{ hot dogs}) + (\$2 \text{ per hamburger} \times 50 \text{ hamburgers}) = \200
2020	$(\$2 \text{ per hot dog} \times 150 \text{ hot dogs}) + (\$3 \text{ per hamburger} \times 100 \text{ hamburgers}) = \600
2021	$(\$3 \text{ per hot dog} \times 200 \text{ hot dogs}) + (\$4 \text{ per hamburger} \times 150 \text{ hamburgers}) = \$1,200$

Calculating Real GDP (base year 2019)

2019	$(\$1 \text{ per hot dog} \times 100 \text{ hot dogs}) + (\$2 \text{ per hamburger} \times 50 \text{ hamburgers}) = \200
2020	$(\$1 \text{ per hot dog} \times 150 \text{ hot dogs}) + (\$2 \text{ per hamburger} \times 100 \text{ hamburgers}) = \350

Prices and Quantities

Year	Price of Hot Dogs	Quantity of Hot Dogs	Price of Hamburgers	Quantity of Hamburgers
2021	$(\$1 \text{ per hot dog} \times 200 \text{ hot dogs}) + (\$2 \text{ per hamburger} \times 150 \text{ hamburgers}) = \500			

Calculating the GDP Deflator

2019	$(\$200/\$200) \times 100 = 100$
2020	$(\$600/\$350) \times 100 = 171$
2021	$(\$1,200/\$500) \times 100 = 240$

To compute total spending in this economy, we multiply the quantities of hot dogs and hamburgers by their prices. In the year 2019, 100 hot dogs are sold at a price of \$1 per hot dog, so expenditure on hot dogs equals \$100. In the same year, 50 hamburgers are sold for \$2 per hamburger, so expenditure on hamburgers also equals \$100. Total expenditure in the economy—the sum of expenditure on hot dogs and expenditure on hamburgers—is \$200. This amount, the production of goods and services valued at current prices, is called **nominal GDP** (the production of goods and services valued at current prices) .

The table shows the calculation of nominal GDP for these three years. Total spending rises from \$200 in 2019 to \$600 in 2020 and then to \$1,200 in 2021. Part of this rise is attributable to the increase in the quantities of hot dogs and hamburgers, and part is attributable to the increase in the prices of hot dogs and hamburgers.

To remove the effect of price changes and obtain a measure of the amount produced, we use **real GDP** (the production of goods and services valued at constant prices) , which is the production of goods and services valued at constant prices. We calculate real GDP by first designating one year as a *base year*. We then use the prices of hot dogs and hamburgers in the base year to compute the value of goods and services in all the years. In other words, the prices in the base year provide the basis for comparing quantities in different years.

Suppose that we choose 2019 to be the base year in our example. We can then use the prices of hot dogs and hamburgers in 2019 to compute the value of goods and services produced in 2019, 2020, and 2021. Table 2 shows these calculations. To compute real GDP for 2019, we multiply the prices of hot dogs and hamburgers in 2019 (the base year) by the quantities of hot dogs and hamburgers produced in 2019. (Thus, for the base year, real GDP always equals nominal GDP.) To compute real GDP for 2020, we multiply the prices of hot dogs and hamburgers in 2019 (the base year) by the quantities of hot dogs and

hamburgers produced in 2020. Similarly, to compute real GDP for 2021, we multiply the prices in 2019 by the quantities in 2021. When we find that real GDP has risen from **\$200** in 2019 to **\$350** in 2020 and then to **\$500** in 2021, we know that the increase is attributable to an increase in the quantities produced because the prices are being held fixed at base-year levels.

To sum up: *Nominal GDP uses current prices to value the economy's production of goods and services. Real GDP uses constant base-year prices to value the economy's production of goods and services.* Because price changes do not affect real GDP, changes in real GDP reflect only changes in the quantities produced. Thus, real GDP measures the economy's production of goods and services.

Our goal in computing GDP is to gauge how well the overall economy is performing. Because real GDP measures the economy's production of goods and services, it reflects the economy's ability to satisfy people's needs and desires. Thus, real GDP is a better gauge of economic well-being than is nominal GDP. When economists talk about the economy's GDP, they usually mean real GDP rather than nominal GDP. And when they talk about growth in the economy, they measure that growth as the percentage change in real GDP from one period to another.

Chapter 23: Measuring a Nation's Income: 23-4a A Numerical Example

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